

VenuePro: The Way Forward

My Vision

I've reviewed our current VenuePro system including the existing codebase, database, and the complete MVP requirements from our test grid. I need to rebuild the system from scratch to properly implement all MVP features while creating a foundation that supports future growth.

Why We Need a Rebuild

Looking at our current system, I've found several issues:

- The database design is inconsistent and hard to maintain
- The code has accumulated technical debt that slows us down
- The current implementation doesn't fully address our MVP requirements
- The user experience needs significant improvement

My Complete MVP Requirements

Based on our test grid and additional requirements listed in our previous documents and correspondences, here's what I'll deliver in the MVP:

1. User Management & Authentication

- Secure login and account creation (using both oauth and the OTP methods discussed)
- Role-based permissions (System Admin, HOD, BA)
- User profile management
- Password reset functionality

2. Room Configuration

- Building creation and configuration
- Room creation with detailed specifications
- Amenity management and assignment
- Room media management (photos)
- Room display functionality

3. Group Management

- Department/group creation and management
- User assignment to groups
- Room access permissions by group
- Ability to disable groups (including handling associated users and bookings)
- Assign groups across multiple buildings

5. Core Booking Engine

- Calendar view with availability display
- Booking creation and management
- Conflict prevention system
- Test calendar for viewing separate rooms simultaneously
- Booking status tracking

6. Advanced Calendar Rules

- Buffer time between bookings
- Minimum notice time for bookings
- Maximum bookings per day (service level)
- Future booking limits
- Daily hour limits
- Custom availability settings at service level
- Calendar overrides (daily break times, holiday settings)
- Ability to disable spaces on holidays

7. Notification System

- Booking confirmations
- Attendee notifications
- Booking change alerts
- Email template management

8. Billing & Invoicing

- Invoicing template creation
- Ability to mark invoices as paid
- Searchable invoicing database
- Define limits and surplus charge configuration
- Usage reporting against set limits
- Live billing and billing behaviors

9. Administration & Reporting

- User management dashboard
- System configuration settings
- Usage tracking and reporting
- Searchable invoice database reporting

My Timeline

Given the expanded scope of the MVP, I'll implement this in focused phases:

Phase 1: Foundation (1 month)

- Implement the improved database schema
- Set up authentication and authorization framework
- Create basic UI templates and components
- Establish development environment and workflow

Phase 2: Core Features (2.5 months)

- User, company, and building management
- Room configuration system
- Group management with cross-building support
- Basic booking engine with conflict prevention
- Room display functionality

Phase 3: Advanced Features (1.5 months)

- Advanced calendar rules implementation
- Buffer system and booking limits
- Holiday management and overrides
- Invoicing and billing system
- Usage limits and surplus charging

Phase 4: Completion & Testing (1 month)

- Notification system
- Reporting mechanisms
- Admin interface completion
- Comprehensive testing of all MVP features
- UI/UX refinement

Technology I'll Use

I'll build the new system with modern, reliable tools:

- **Backend:** PHP (Laravel 10)
- **Database:** PostgreSQL with the improved schema design
- **Frontend:** Vue with Tailwind CSS for responsive design

Feature Implementation Approach

Group Management

I'll build the group system to:

- Allow complete group deactivation (with cascading effects on users/bookings)
- Support cross-building group assignment
- Handle duplicate room names across buildings

Calendar Rules

I'll implement a flexible rules engine that supports:

- Buffer time between bookings
- Configurable notice periods
- Daily booking limits
- Service-level availability settings
- Holiday and break time management

Billing System

I'll create a comprehensive billing module that:

- Generates professional invoices from templates
- Tracks payment status
- Implements configurable charging rules
- Provides searchable billing history
- Calculates surplus charges automatically

Testing Strategy

For the expanded MVP, I'll implement a rigorous testing approach:

1. Test calendar functionality with multiple rooms simultaneously
2. Verify group management across multiple buildings
3. Validate all calendar rules and edge cases
4. Test billing calculations and invoice generation

Major Milestones

Milestone	When
Foundation Complete	Month 1
Core Features Ready	Month 3.5
Advanced Features Ready	Month 5
MVP Ready for Testing	Month 5.5
MVP Launch	Month 6

Next Steps

1. Review this complete MVP plan
2. Approve the proposed database schema improvements
3. Set up the development environment and source control
4. Begin foundation phase development
5. Establish weekly progress reviews

I'll continuously refer to our original test grid and previous documents throughout development to ensure we're delivering all required MVP features.